



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22AI503/22AM503 MACHINE LEARNING

UNIT II – DATA PREPROCESSING

Basic Statistical Description of Data

1. Introduction

1.1 Definition

Basic statistical description of data refers to the use of descriptive statistics to summarise, visualise, and understand the characteristics of a dataset before applying machine learning algorithms. It includes measures of central tendency, dispersion, shape, and relationships between attributes.

Definition:

Descriptive Statistics provides a quantitative summary of a dataset's main features.

It is the first exploratory step in any ML or data mining project.

Helps identify outliers, data quality issues, distributions, and correlations.

1.2 Importance in Machine Learning

Statistical profiling before modelling helps identify skewed distributions that require transformation, outliers that distort model training, missing values patterns, correlation structures that guide feature selection, and appropriate preprocessing strategies per attribute type.

1.3 Role in Data Preprocessing Pipeline

Statistical description is performed in the Exploratory Data Analysis (EDA) phase. It informs decisions about normalization (when variance is large), outlier handling (when IQR-based anomalies exist), binning (when continuous attrs need discretization), and feature selection (using correlation matrices).

2. Learning Outcomes

Understand

- Measures of central tendency: mean, median, mode, and their sensitivity to outliers
- Measures of dispersion: range, variance, standard deviation, IQR
- Distribution shape: skewness and kurtosis

Apply

- Compute five-number summary and construct boxplots for outlier detection
- Calculate Pearson and Spearman correlation coefficients
- Use pandas describe() and seaborn/matplotlib for EDA

Analyze

- Interpret skewness and kurtosis to choose appropriate transformations
- Distinguish between positive and negative correlation and their implications

- Identify outliers using Z-score and IQR methods

3. Detailed Technical Concepts

3.1 Measures of Central Tendency

3.1.1 Mean (Arithmetic Average)

The mean is the sum of all values divided by the count of values:

$$\bar{x} = (1/n) \times \sum x_i \text{ for } i = 1 \text{ to } n$$

Properties: Sensitive to outliers. If one value is extremely large, the mean shifts significantly. Best used when data is approximately symmetric and outlier-free.

Weighted Mean: $\bar{x}_w = \sum(w_i \times x_i) / \sum w_i$ — used when observations have different importances.

3.1.2 Median

The median is the middle value when data is sorted. For even n: median = average of $n/2$ and $(n/2+1)$ values. Robust to outliers — preferred when data is skewed or contains extreme values.

Example: [2, 3, 5, 7, 8, 100] → Median = $(5+7)/2 = 6$ (unaffected by 100)

3.1.3 Mode

The mode is the most frequently occurring value. Data may be unimodal (one mode), bimodal, or multimodal. Mode is the only measure applicable to nominal attributes.

Example: [3, 3, 5, 7, 8, 8, 8] → Mode = 8 (appears 3 times)

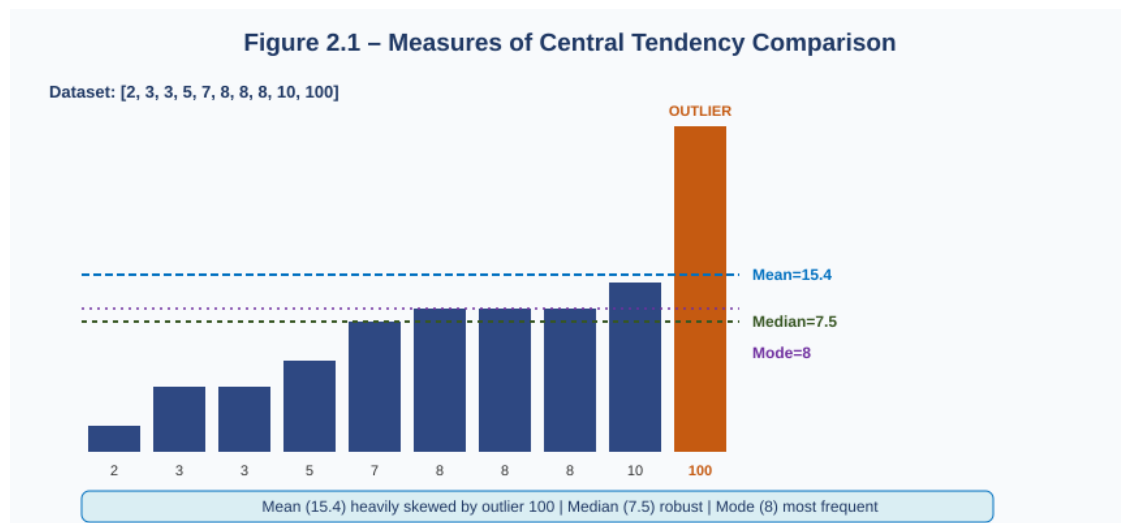


Figure 2.1 – Measures of Central Tendency Comparison on Skewed Data

3.2 Measures of Dispersion / Spread

3.2.1 Range

Range = Max – Min. Simple but sensitive to outliers. Example: [2, 5, 8, 100] → Range = 98 (inflated by outlier 100).

3.2.2 Variance and Standard Deviation

Population Variance: $\sigma^2 = (1/n) \times \sum (x_i - \bar{x})^2$

Sample Variance: $s^2 = (1/(n-1)) \times \sum (x_i - \bar{x})^2$ (Bessel's correction for unbiased estimate)

Standard Deviation: $\sigma = \sqrt{\sigma^2}$ (same units as original data, more interpretable)

Low SD: data points clustered near the mean. High SD: data spread widely.

3.2.3 Interquartile Range (IQR)

IQR = $Q3 - Q1$ where $Q1 = 25\text{th percentile}$, $Q3 = 75\text{th percentile}$.

IQR is robust to outliers. Used in boxplots. Outlier fences: $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$.

3.2.4 Percentiles and Quartiles

The p-th percentile is the value below which p% of observations fall. The five-number summary (Min, $Q1$, $Q2$, $Q3$, Max) gives a robust statistical profile of any numeric attribute.

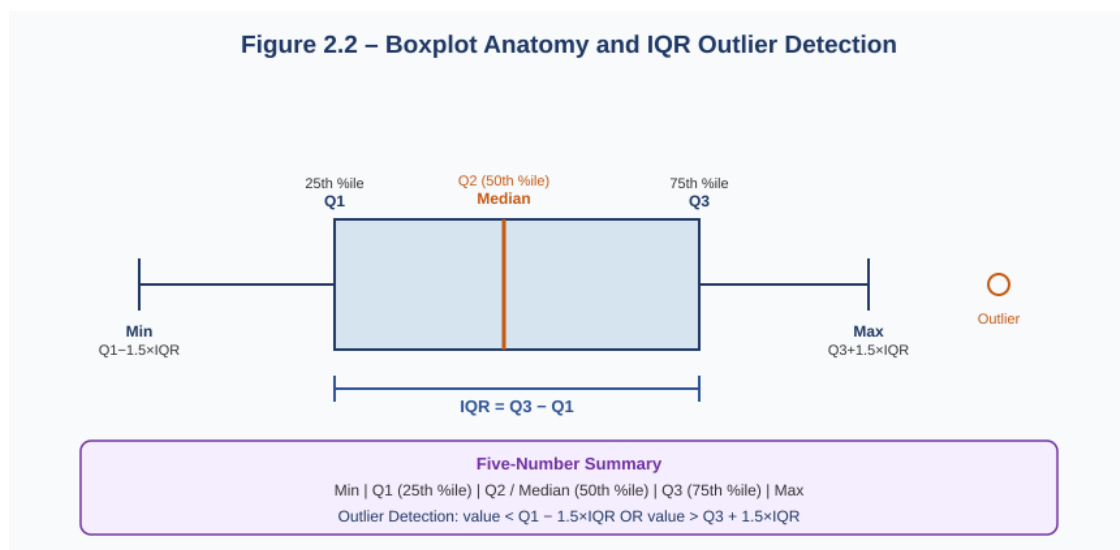


Figure 2.2 – Boxplot Anatomy and IQR-Based Outlier Detection

3.3 Distribution Shape: Skewness and Kurtosis

3.3.1 Skewness

Skewness measures asymmetry of the distribution. Pearson's coefficient: Skewness = $3 \times (\text{Mean} - \text{Median}) / \text{StdDev}$.

- Skewness = 0: Symmetric distribution (Normal)
- Skewness > 0: Right-skewed / Positive skew (tail extends right; Mean > Median > Mode). Common in income, house prices.
- Skewness < 0: Left-skewed / Negative skew (tail extends left; Mean < Median < Mode). Common in exam scores near ceiling.

Rule of thumb: $|\text{Skewness}| > 1$ indicates high skewness requiring transformation (log, square root, Box-Cox).

3.3.2 Kurtosis

Kurtosis measures the "peakedness" and tail heaviness of a distribution. Excess Kurtosis = Kurtosis - 3.

- Mesokurtic (Excess Kurtosis = 0): Normal distribution — baseline reference
- Leptokurtic (Excess Kurtosis > 0): Sharper peak, heavier tails — more outliers than normal (e.g., financial returns)
- Platykurtic (Excess Kurtosis < 0): Flatter peak, lighter tails — fewer outliers (e.g., uniform distribution)

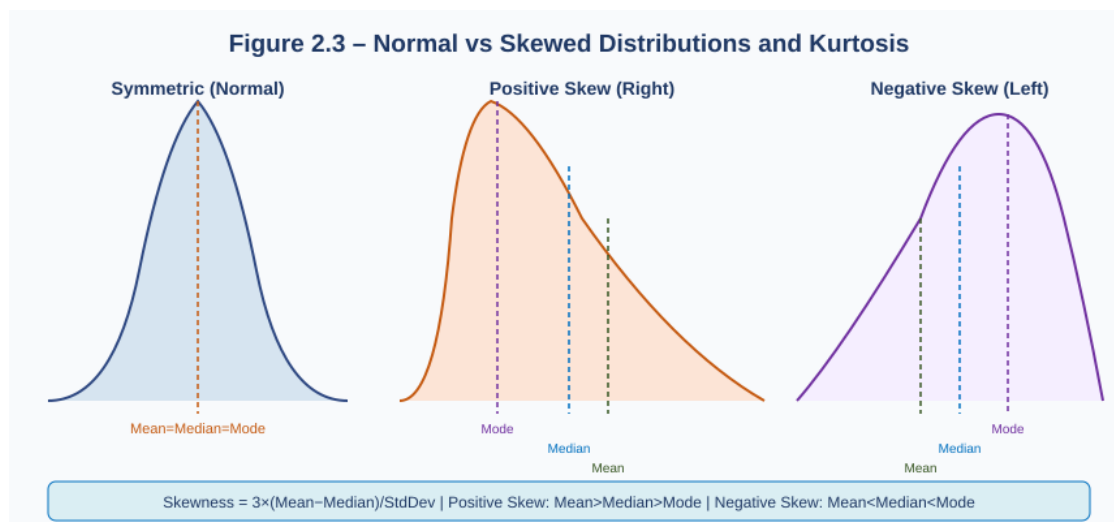


Figure 2.3 – Normal, Positive Skew, and Negative Skew Distribution Shapes

3.4 Correlation Analysis

3.4.1 Pearson Correlation Coefficient (r)

Measures linear relationship between two numeric attributes X and Y:

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \times \sum(y_i - \bar{y})^2]}} \quad \text{where } -1 \leq r \leq 1$$

- $r = +1$: Perfect positive linear correlation
- $r = 0$: No linear correlation (may have non-linear relationship)
- $r = -1$: Perfect negative linear correlation
- $|r| > 0.7$: Strong correlation; 0.3–0.7: Moderate; < 0.3: Weak

Limitation: Pearson measures only linear relationships and is sensitive to outliers.

3.4.2 Spearman Rank Correlation

Non-parametric measure using ranks instead of raw values. Suitable for ordinal data and non-linear monotonic relationships:

$$\rho = 1 - \frac{6 \times \sum d_i^2}{(n \times (n^2 - 1))} \quad \text{where } d_i = \text{rank}(x_i) - \text{rank}(y_i)$$

3.4.3 Covariance

$\text{Cov}(X, Y) = (1/(n-1)) \times \sum[(x_i - \bar{x})(y_i - \bar{y})]$. Positive covariance: X and Y increase together. Negative: one increases while other decreases. Note: Covariance is scale-dependent; Pearson r is normalised.

3.5 Graphical Summaries — Key EDA Visualisations

Plot Type	Purpose	Applicable Attribute	Key Insight
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Histogram	Show frequency distribution	Numeric (interval/ratio)	Detect skewness, modality, spread
Boxplot	Show 5-number summary + outliers	Numeric	Outlier detection, IQR visualisation
Scatter Plot	Show pairwise relationship	2 Numeric attrs	Correlation direction and strength
Bar Chart	Show category frequencies	Nominal / Ordinal	Class imbalance, dominant categories
Q-Q Plot	Check normality assumption	Numeric	Points on diagonal = normal distribution
Heatmap	Show correlation matrix	All numeric pairs	Identify highly correlated attr groups
KDE Plot	Smooth density estimate	Numeric (continuous)	Better than histogram for smooth view
Violin Plot	Distribution + boxplot combined	Numeric	Compare distributions across groups

3.6 Algorithms and Techniques

Z-score Outlier Detection: $z = (x - \mu) / \sigma$. Values with $|z| > 3$ are typically flagged as outliers (assuming normal distribution). More conservative than IQR method.

Modified Z-score (robust): $M = 0.6745 \times (x_i - \text{Median}) / \text{MAD}$ where MAD = Median Absolute Deviation. Preferred for non-normal distributions.

4. EDA Statistical Description Workflow

Step	Action	Python / pandas Code	Output
1	Load data and inspect schema	<code>df.dtypes, df.shape, df.head()</code>	Column names, types, row count
2	Generate summary statistics	<code>df.describe(include='all')</code>	Mean, std, min, Q1, Q2, Q3, max
3	Check missing values	<code>df.isnull().sum() / len(df) * 100</code>	Missing % per column
4	Check categorical distributions	<code>df[col].value_counts(normalize=True)</code>	Category frequencies
5	Compute skewness and kurtosis	<code>df.skew(), df.kurtosis()</code>	Shape metrics per numeric column
6	Correlation matrix	<code>df.corr(method='pearson')</code>	Pearson r matrix
7	Outlier detection (IQR)	<code>Q1=df.quantile(0.25); Q3=df.quantile(0.75); IQR=Q3-Q1</code>	Outlier mask
8	Visualise with boxplots/histograms	<code>sns.boxplot(); plt.hist()</code>	Distribution shapes

5. Industry Tools and Technologies

5.1 Commercial Tools

Tool	Vendor	Statistical Features	Visualization
SPSS Statistics	IBM	Full descriptive stats, hypothesis testing	Built-in charts, Q-Q plots
SAS Base	SAS	PROC UNIVARIATE, PROC CORR	SGPLOT, box/scatter/hist
Tableau	Tableau	Reference lines, distribution overlays	Interactive dashboards
Power BI	Microsoft	DAX statistical functions	Violin plots via R/Python visuals
JMP	SAS	Distribution explorer, correlation matrix	Interactive EDA platform

5.2 Open-Source Tools

- pandas: df.describe(), df.corr(), df.skew(), df.kurtosis()
- scipy.stats: pearsonr(), spearmanr(), normaltest(), shapiro()
- seaborn: boxplot(), heatmap(), pairplot(), violinplot(), kdeplot()
- matplotlib: hist(), scatter(), bar()
- ydata-profiling (formerly pandas-profiling): auto EDA report with stats + plots
- R: summary(), cor(), boxplot(), shapiro.test(), ggplot2

6. Real-World Applications

Domain	Statistical Use	Key Stat Metric	Action Taken
Credit Scoring	Describe loan applicant income distribution	Skewness = +2.3 (right-skewed)	Log-transform income before modelling
Manufacturing QC	Monitor sensor readings for anomalies	Z-score > 3 for vibration sensor	Flag unit for inspection
Healthcare	Compare blood glucose across age groups	ANOVA + mean/std per group	Identify diabetic risk segments
Retail	Analyse sales distribution per product	IQR-based boxplot outlier detection	Remove data entry errors in sales
Finance	Model stock return distributions	Excess kurtosis = +4.2 (leptokurtic)	Use heavy-tail models (Student-t)

7. Advantages and Limitations

Advantages	Limitations
Fast and computationally inexpensive	Summary stats can hide complex multivariate patterns
Identifies outliers before training	Pearson correlation misses non-linear relationships

Guides transformation decisions (log, sqrt)	Skewness/kurtosis less meaningful for small n (<30)
Correlation matrix enables feature selection	High correlation does not imply causation
Boxplots provide intuitive outlier visualisation	Mean/SD misleading for multimodal distributions

8. Case Study

CASE STUDY

Case: Boston Housing Dataset — Skewness-Driven Feature Engineering

Problem: A regression team building a house price prediction model found $R^2 = 0.68$ using raw features in Linear Regression.

Dataset: 506 records | 13 numeric features | Target: MEDV (median house value in \$1000s)

Statistical Analysis:

- CRIM (crime rate): Skewness = +5.2 (extreme right-skew)
- DIS (distance to employment): Skewness = +2.1
- LSTAT (% lower-status population): Skewness = +0.9
- MEDV (target): Skewness = +1.1

Findings: Pearson correlation matrix revealed LSTAT ($r = -0.74$) and RM (rooms, $r = +0.70$) are the strongest predictors. Outlier analysis via boxplot found 8 records with $CRIM > Q3 + 1.5 \times IQR$.

Action Taken: Log-transform applied to CRIM, DIS, LSTAT. Outliers removed (8 records). Re-run regression: R^2 improved from 0.68 to 0.82.

Conclusion: Statistical description (skewness check, correlation matrix, IQR-outlier removal) guided targeted preprocessing, improving predictive accuracy by 14 percentage points.

9. Scenario-Based Learning

SCENARIO 1

SCENARIO 1: Skewed Income Feature in Credit Risk Model

Background: A fintech company trains a credit default prediction model. The "annual_income" feature shows extreme right skew (skewness = +4.7) with income values ranging from \$10K to \$8M.

Technical Environment: Python, XGBoost, 500,000 loan records, sklearn pipeline.

Problem Statement: XGBoost trained on raw income shows poor performance on low-income applicants (recall = 0.31 for high-risk segment).

Available Evidence: `df['annual_income'].skew()` = 4.7. Boxplot shows 2,300 outliers above $Q3 + 1.5 \times IQR$.

Investigation: Check mean vs median: Mean = \$82K, Median = \$52K (gap indicates severe skew).

Analysis: Log transformation compresses the income scale, making the distribution approximately normal.

Findings: After log transform, skewness reduced to +0.3. Recall for high-risk segment improved to 0.74.

Overall AUC improved from 0.81 to 0.87.

Recommended Solution: Apply `np.log1p()` to income. Cap outliers at 99th percentile before transformation. Validate skewness post-transform is within (-0.5, +0.5).

SCENARIO 2

SCENARIO 2: Multicollinear Features Detected via Correlation Matrix

Background: A data scientist building a house price model has 30 features including square footage, number of rooms, number of bathrooms, and total built area.

Technical Environment: Linear Regression, sklearn, pandas, matplotlib.
 Problem Statement: Model produces unexpected negative coefficients for rooms and square footage, making the model uninterpretable.
 Available Evidence: `df.corr()` shows: sqft vs rooms = 0.96, sqft vs bathrooms = 0.91, rooms vs bathrooms = 0.89 (all highly correlated).
 Investigation: VIF (Variance Inflation Factor) computed: $VIF(sqft) = 14.2$, $VIF(rooms) = 11.8$ (both > 10 indicating multicollinearity).
 Analysis: Multicollinear features inflate coefficient variance, causing sign flips and instability.
 Findings: Dropping rooms and bathrooms (retaining sqft as representative) reduced VIF to 2.1. R^2 minimally decreased (0.84 → 0.83) but coefficient signs became correct.
 Recommended Solution: Use correlation matrix + VIF to detect multicollinearity. Drop or merge correlated features. Alternatively apply PCA for dimensionality reduction.

SCENARIO 3

SCENARIO 3: Outlier Contamination in Medical Glucose Data

Background: A diabetes prediction model was trained on a medical dataset. During evaluation, the model consistently predicted "diabetic" for healthy patients.
 Technical Environment: Random Forest, PIMA Indians Diabetes dataset, scikit-learn.
 Problem Statement: Boxplot of glucose shows values of 0 for 5 records — physiologically impossible (minimum normal glucose ~40 mg/dL).
 Available Evidence: `df[df['glucose']==0].count()` = 5. Also: `df['bmi']==0` for 11 records. These are missing values encoded as 0.
 Investigation: `df.describe()` shows `min(glucose) = 0`, which is physically invalid. IQR check: $Q1=99$, $Q3=140$, $IQR=41$; fence = $99 - 61.5 = 37.5$ (so 0 is outside).
 Analysis: Zero values are not true zeros but represent missing data. Treating them as real values biases the mean downward ($118 \rightarrow 114$ with zeros included).
 Findings: After replacing 0s with NaN and imputing with median (117 mg/dL), model AUC improved from 0.79 to 0.83.
 Recommended Solution: Always verify physically impossible values for medical features using domain knowledge. Use `df[col].replace(0, np.nan)` followed by median imputation or model-based imputation.

10. Common Errors and Troubleshooting

Error	Cause	Impact on Model	Fix
Using mean on skewed data for imputation	Ignoring skewness in EDA	Biased imputation pulls values toward outliers	Use median imputation for skewed attrs
Assuming Pearson $r = 0$ means independence	Misunderstanding correlation	Miss non-linear relationships (e.g., quadratic)	Use Spearman or MIC (Maximal Info Coeff)
Ignoring zero/impossible values as outliers	Treating encoding errors as real data	Model learns from erroneous patterns	Domain check + replace with NaN
Using raw variance without standardization	Different units/scales across features	Distance-based models biased toward high-range feats	Apply StandardScaler or MinMaxScaler
Reporting only mean and SD for bimodal data	Incomplete statistical profiling	Hides mixture distributions (two populations)	Always visualise histograms; use KDE + check modality

11. Key Terminologies

Term	Definition
Mean	Arithmetic average of all values: $(\sum x_i)/n$. Sensitive to outliers.
Median	Middle value in sorted data. Robust to outliers.
Mode	Most frequently occurring value. Can be applied to nominal attributes.
Variance (s^2)	Average squared deviation from mean: $\sum (x_i - \bar{x})^2 / (n-1)$
Standard Deviation	Square root of variance. Same unit as original data.
IQR	Interquartile Range = $Q3 - Q1$. Robust spread measure.
Skewness	Asymmetry of distribution. Positive = right tail longer.
Kurtosis	Tail heaviness. Excess kurtosis > 0 = heavier tails than normal.
Percentile	Value below which a given percentage of observations fall.
Pearson r	Linear correlation coefficient between -1 and $+1$.
Covariance	Measure of joint variability of two attributes.
Z-score	Standardised score: $(x - \mu)/\sigma$. $ z > 3$ flags outliers.
Five-Number Summary	Min, Q1, Q2, Q3, Max — compact dataset profile.
KDE	Kernel Density Estimate: smooth continuous approximation of a histogram.

12. Topic Summary

UNIT II – Topic 2

BASIC STATISTICAL DESCRIPTION OF DATA — SUMMARY

- Central Tendency: Mean (sensitive to outliers), Median (robust), Mode (nominal data). Use median for skewed distributions.
- Dispersion: Range (simple), Variance/SD (standard), IQR (robust). IQR used in boxplot outlier fences: $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$.
 - Shape: Skewness (asymmetry) and Kurtosis (tail heaviness) guide transformation choices.
 - Correlation: Pearson r for linear numeric relationships; Spearman ρ for ordinal/non-linear.
 - Five-Number Summary (Min, Q1, Median, Q3, Max) is the standard boxplot construction.
 - Key EDA plots: Histogram, Boxplot, Scatter, Correlation Heatmap, Q-Q Plot.
 - Z-score ($|z| > 3$) and IQR methods detect outliers; physiologically impossible values need domain validation.
 - Tools: pandas describe(), scipy.stats, seaborn, ydata-profiling, SPSS, SAS PROC UNIVARIATE.
- Statistical profiling directly informs normalization, outlier removal, transformation, and feature selection decisions.